

Financial Market Regime Detection Using Hidden Markov Chains

A project report
submitted by

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1 Introduction

1.1 History of Markov Chains

The development of Markov chains began with the pioneering work of Russian mathematician Andrey Markov in the early 20th century (circa 1906), who initially studied these stochastic processes in the context of analyzing sequences of letters in Russian literature [7]. A Markov chain represents a sequence of possible events where the probability of each event depends solely on the state attained in the previous event—a property known as the Markov property. While traditional Markov chains operate with directly observable states, Hidden Markov Models (HMMs) emerged as a natural extension in the late 1960s through the work of Leonard E. Baum and others. In an HMM, the system being modeled is assumed to be a Markov process with unobservable (hidden) states that generate observable outputs. This powerful framework has proven instrumental in numerous applications including speech recognition [3], bio-informatics [4], natural language processing [3], and financial modeling [5], where underlying processes cannot be directly observed but manifest through measurable outputs.

1.2 Research Structure

This report presents a comprehensive analysis of Hidden Markov Models (HMMs), encompassing both their implementation to macro-economic variables and empirical results. The document is structured to first establish the mathematical foundations by defining the derivations, then illustrate the underlying concepts via examples of both standard Markov chains and their hidden variants. Subsequently, the paper examines the specific applications of HMMs within the financial domain, detailing the methodologies employed and analyzing the resulting outcomes. Through this structured approach, the reader will be provided with a comprehensive understanding of Hidden Markov Models, from their theoretical foundations to practical implementations, demonstrating their versatility and effectiveness across the financial domain.

2 Mathematical Derivations

2.1 Markov Chains

A Markov chain is a stochastic process that satisfies the Markov property, which stipulates that the future state depends only on the current state and not on the sequence of events that preceded it. Formally, a Markov chain is defined as follows:

Let $X_t : t = 0, 1, 2, \dots$ be a sequence of random variables with state space S . This sequence is a Markov chain if for any $t \geq 0$ and any states $i, j \in S$:

$$P(X_{t+1} = j | X_t = i, X_{t-1} = i_{t-1}, \dots, X_0 = i_0) = P(X_{t+1} = j | X_t = i) \quad (1)$$

This conditional probability is called the transition probability and is typically denoted as:

$$p_{ij} = P(X_{t+1} = j | X_t = i) \quad (2)$$

2.1.1 Key Parameters and Components

State Space (S): The set of all possible states the system can occupy. For a finite Markov chain, $S = 1, 2, \dots, N$.

Transition Matrix (P): An $N \times N$ matrix where each element p_{ij} represents the probability of transitioning from state i to state j in one step:

$$P = \begin{pmatrix} p_{11} & p_{12} & \cdots & p_{1N} \\ p_{21} & p_{22} & \cdots & p_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ p_{N1} & p_{N2} & \cdots & p_{NN} \end{pmatrix} \quad (3)$$

The transition matrix satisfies:

$$p_{ij} \geq 0 \text{ for all } i, j \in S$$

$$\sum_{j \in S} p_{ij} = 1 \text{ for all } i \in S$$

Initial State Distribution (δ): A vector where $\delta_i = P(X_0 = i)$ represents the probability that the Markov chain starts in state i :

$$\delta = (\delta_1, \delta_2, \dots, \delta_N)$$

with $\delta_i \geq 0$ and $\sum_{i=1}^N \delta_i = 1$.

n-step Transition Probability ($p_{ij}^{(n)}$): The probability of transitioning from state i to state j in exactly n steps:

$$p_{ij}^{(n)} = P(X_{t+n} = j | X_t = i) \quad (4)$$

These can be computed using the n -th power of the transition matrix: P^n .

Stationary Distribution (μ): A probability distribution that remains unchanged by the transition matrix:

$$\mu P = \mu \quad (5)$$

A Markov chain is fully characterized by its state space S , transition matrix P , and initial distribution δ . These parameters are essential for analyzing the behavior of the system over time and for making predictions about future states based on the current state.

2.1.2 Markov Chain Example

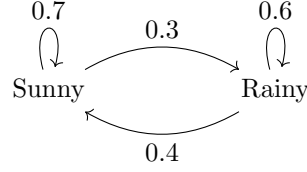
Let's consider a simple Markov chain model for daily weather patterns with two states: "Sunny" and "Rainy".

$$S = \{Sunny, Rainy\}$$

Suppose we have observed the following transition probabilities:

$$P = \begin{pmatrix} p_{SS} & p_{SR} \\ p_{RS} & p_{RR} \end{pmatrix} = \begin{pmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{pmatrix}$$

The transition matrix visualized:



This primitive example highlights how each state has an associated probability with moving to another. This behavior is the core of a Markov Chain.

2.2 Hidden Markov Chains

A Hidden Markov Model (HMM) extends the concept of Markov chains by introducing a layer of unobservable (hidden) states. While the state sequence itself follows Markov properties, we can only observe outputs generated by these hidden states rather than the states themselves.

2.2.1 Key Parameters and Components

Hidden State Space (S): A set of N possible hidden states, denoted as $S = s_1, s_2, \dots, s_N$, which follow the Markov property but are not directly observable.

Observation Space (O): A set of M possible observations, denoted as $O = o_1, o_2, \dots, o_M$, which are the visible outputs of the system.

Transition Probability Matrix (P): An $N \times N$ matrix where each element p_{ij} represents the probability of transitioning from hidden state s_i to hidden state s_j :

$$P = \begin{pmatrix} p_{11} & p_{12} & \cdots & p_{1N} \\ p_{21} & p_{22} & \cdots & p_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ p_{N1} & p_{N2} & \cdots & p_{NN} \end{pmatrix} \quad (6)$$

where $p_{ij} = P(q_{t+1} = s_j | q_t = s_i)$ and q_t denotes the state at time t .

Emission Probability Matrix (B): An $N \times M$ matrix where each element $b_i(k)$ represents the probability of observing o_k while in hidden state s_i :

$$B = \begin{pmatrix} b_1(1) & b_1(2) & \cdots & b_1(M) \\ b_2(1) & b_2(2) & \cdots & b_2(M) \\ \vdots & \vdots & \ddots & \vdots \\ b_N(1) & b_N(2) & \cdots & b_N(M) \end{pmatrix} \quad (7)$$

where $b_i(k) = P(v_t = o_k | q_t = s_i)$ and v_t denotes the observation at time t .

Initial State Distribution (δ): A vector where δ_i represents the probability that the Markov chain starts in hidden state s_i :

where

$$\delta_i = P(q_1 = s_i) \quad (8)$$

2.2.2 Hidden Markov Chain Example

Let's consider an HMM for financial markets with:

$$S = \{Bear, Bull\}$$

$$O = \{Up, Down, Flat\}$$

$$P = \begin{pmatrix} p_{Bear,Bear} & p_{Bear,Bull} \\ p_{Bull,Bear} & p_{Bull,Bull} \end{pmatrix} = \begin{pmatrix} 0.8 & 0.2 \\ 0.3 & 0.7 \end{pmatrix}$$

$$B = \begin{pmatrix} b_{Bear}(Up) & b_{Bear}(Flat) & b_{Bear}(Down) \\ b_{Bull}(Up) & b_{Bull}(Flat) & b_{Bull}(Down) \end{pmatrix} = \begin{pmatrix} 0.2 & 0.3 & 0.5 \\ 0.6 & 0.3 & 0.1 \end{pmatrix}$$

This means:

In a bear market: prices go down with 50% probability, stay flat with 30%, and rise with only 20% In a bull market: prices go up with 60% probability, stay flat with 30%, and fall with only 10%

3 Research

3.1 Approach

The research and experiments performed are largely based off the white-paper called *RISKS* by Nyugen and Nygen [6]. The reason for this was to have an accuracy benchmark for an existing, proven model which can determine various states using an HMM. The experiments in both *RISKS* and this research analyze the following four macro-economic variables: S&P 500 Index (S&P500), Consumer Price Index of Total Urban Consumers (CPIAUCNS), Industrial Production Index (INDPRO), and the Total Market Volatility Index (VIX).

Various python scripts were written to create separate HMMs applied to each aforementioned macro-economic variable. The following algorithms were used to produce outputs of a given HMM:

- Forward algorithm
 - Calculates the probability of observing a specific sequence of states.
- Backward algorithm
 - Computes probability of future observations given a current state.
- Viterbi algorithm
 - Determines the most likely sequence of hidden states for a given observation sequence.
- Baum-Welch algorithm
 - Learns the optimal emission probabilities from observation data through iterative expectation-maximization.

All four macro-economic variables implore a two state HMM. The description of the index, the states, and the data used are all listed below.

3.2 S&P 500 Index

The S&P 500 is a stock market index that measures the performance of the 500 largest companies listed on stock exchanges in the United States. As a capitalization-weighted index that includes companies across various sectors including technology, healthcare, finance, and consumer discretionary, the S&P

500 captures approximately 80% of the available market capitalization, making it a critical benchmark for investors and a key indicator of broader market movements.

The states which are used in this HMM are *bull* and *bear*. The data has been harvested from Yahoo finance (<https://finance.yahoo.com>).

3.3 CPIAUNCS

The CPIAUCNS is a comprehensive measure produced by the U.S. Bureau of Labor Statistics that tracks the average change over time in prices paid by urban consumers for a market basket of consumer goods and services. This economic indicator serves as the primary gauge of inflation in the United States economy, reflecting price changes for items ranging from housing and transportation to food, medical care, and education, thereby providing valuable insights into the purchasing power of the U.S. dollar.

The states which are used in this HMM are *Inflation* and *Deflation*. The data has been harvested from Bureau of Labor Statistics, U.S. Department of Labor (<http://www.bls.gov/cpi/>).

3.4 INDPRO

The INDPRO is an economic indicator produced by the Federal Reserve Board that measures real output in the manufacturing, mining, electric, and gas utilities sectors of the U.S. economy. This vital metric quantifies changes in the physical volume of production across various industrial facilities, providing insights into the health of the U.S. industrial sector and serving as a leading indicator of economic expansions and contractions, with particular sensitivity to business cycles and inflationary pressures that potentially impact financial markets.

The states which are used in this HMM are *Growth* and *Recession*. The data has been harvested from the Federal Reserve System (<http://www.federalreserve.gov/>).

3.5 VIX

The VIX represents the market's expectations for volatility, calculated from the implied volatility of S&P 500 index options. The VIX typically rises during periods of financial stress and market uncertainty and declines during periods of market stability, providing investors with a quantitative measure of market

risk sentiment that serves as an input for trading strategies, risk management frameworks, and macroeconomic analysis.

The states which are used in this HMM are *High Volatility* and *Low Volatility*. The data has been harvested from the Chicago Board of Options Exchange (<http://www.cboe.com/>).

3.6 Data Summary

Variable	Data Description	State 1	State 2
S&P 500	MoM % change	Bull	Bear
CPIAUNCS	YoY % change	Inflation	Deflation
INDPRO	MoM % change	Growth	Recession
VIX	MoM % change	High Volatility	Low Volatility

Table 1: Variable Descriptions and State Classifications

Please note that all data has been log-adjusted.

4 Results

4.1 S&P 500

The graph below represents the detected regimes from 1994-2024, where the shaded and white portions of the graph are bear and bull, respectively.

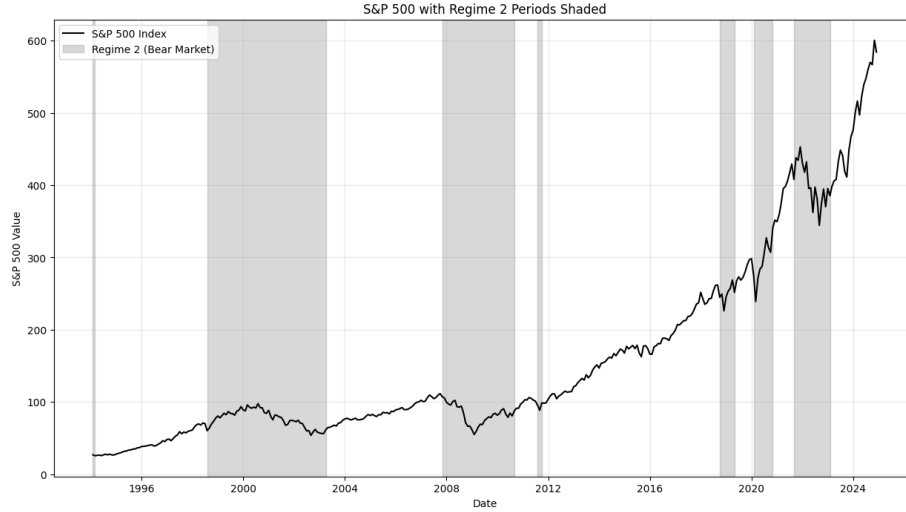


Figure 1: S&P 500 performance over a 20-year period.

The model performs quite well when detecting states. It is clearly shown that during Y2K, dot-com bubble, housing crisis of '08, and covid, we get significant readings of being in a bear market. This correlates to the events which occurred during the economy during those times. As shown, this implementation of HMM can effectively identify market regimes in equity markets, allowing investors to adjust their strategies according to the estimated current state. It's important to note that there is no universal, formal definition for a bear market, but is widely accepted that it is a prolonged period of price decline (typically referred to as a 20% decline) in conjunction with a fearful investor sentiment [1].

4.2 CPIAUNCS

Inflationary periods for an economy are the norm. As humans, we have a remarkable characteristic among us for quantifying the value of some product or relationship therein. Therefore, it is standard to have inflationary periods. The graph below highlights this as having mostly times of inflationary periods over the last twenty years. The shaded area represents periods of inflation, while the white are periods of deflation.

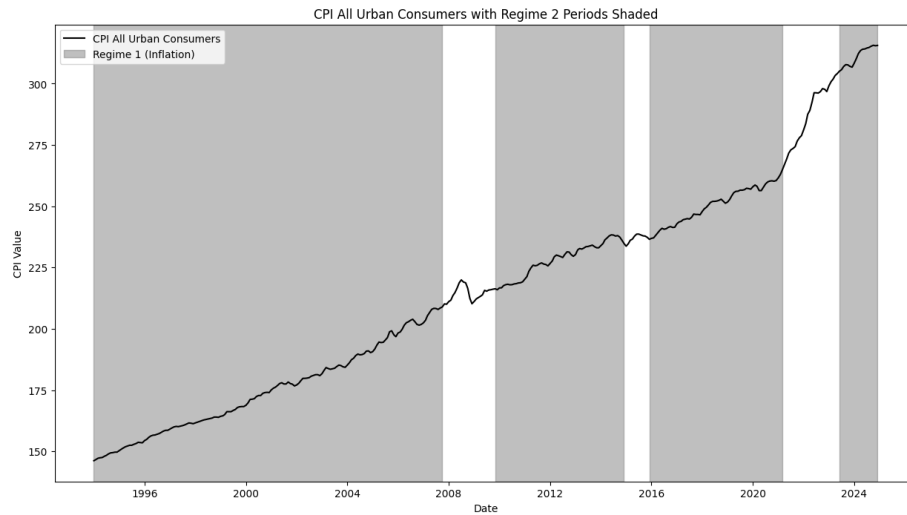


Figure 2: CPIAUNCS readings over a 20-year period.

To corroborate these results, below is a list of deflationary periods from 1994-2024.

- 2008 - 2009 Housing Crisis
- 2015 Oil Price Collapse
- Covid Era, March - May 2020

The model does an excellent job detecting periods of deflation. However, during a sharp increase of inflation (early 2022 - late 2023), the model produces an erroneous reading. Perhaps this is because the typical rate of inflation is not followed during this time period, so the hidden model has to fit the abnormality into the other state.

4.3 INDPRO

Recall the INDPRO signifies the real output in the manufacturing, mining, electric, and gas utilities sectors of the U.S. economy. This macro-economic variable was the toughest to get an accurate reading. This is likely due to the nature of how production can change due to socio-political movements and the intricacies of supply and demand. It's important to note the time period of 2007 - 2024 does not have much change in the max and mins of production output. Therefore, even if the model is not seeing a consistent increase in production, it must adapt to the patterns at hand. Let's explore how it performs; the grey shaded area represents recession, while the white is growth.

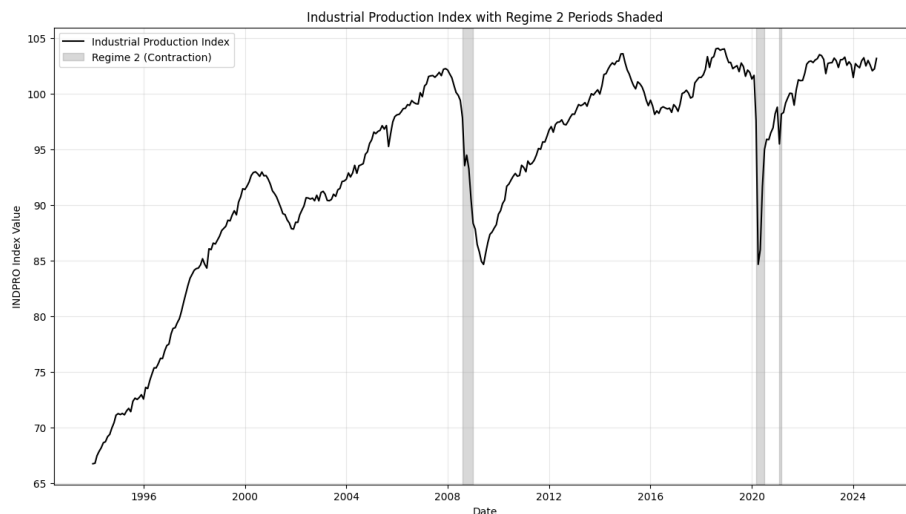


Figure 3: INDPRO readings over a 20-year period.

The following list are time periods of economic recessions in the U.S.:

- Dot-Com Recession (March 2001 - Nov 2001) (8 months)
- The Great Recession (Dec 2007 - June 2009) (18 months)
- Covid 19 Recession (Feb 2020 - April 2020) (2 months)

While the model doesn't accurately capture the dotcom recession, the other two periods of recession are accurately depicted. Perhaps a reason for the missed reading is because many time periods must pass through the HMM to produce an accurate transition matrix. In practice, this is often circumvented by ignoring early outputs of the model or by back-filling data such that the starting period of the use case is considered accurate, neither of which was done in this research.

4.4 VIX

Market volatility can happen independent of a given regime. There are many reasons which a time period could be considered highly volatile which include: IPO listings, scandals, macro economic regime shifts, political change, bankruptcies, etc. For these reasons, the graph below produces many readings of a high-volatility period, which is the grey shaded area. The white area represents periods of low volatility.

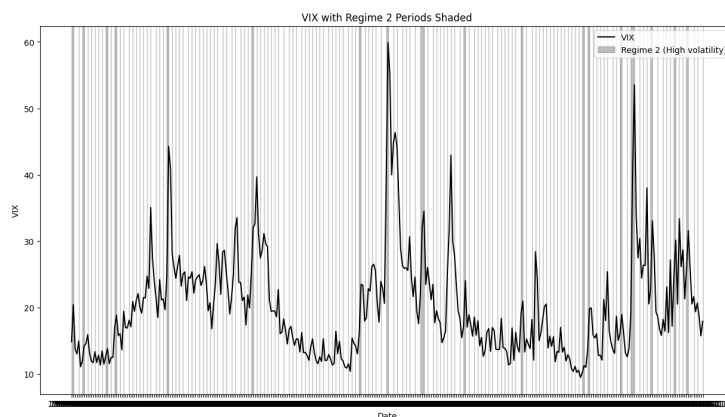


Figure 4: VIX readings over a 20-year period.

There are so many readings of high-volatility that we will need to zoom in and have a closer look to see what is occurring. The next page shows the VIX readings during the housing crisis of '08.

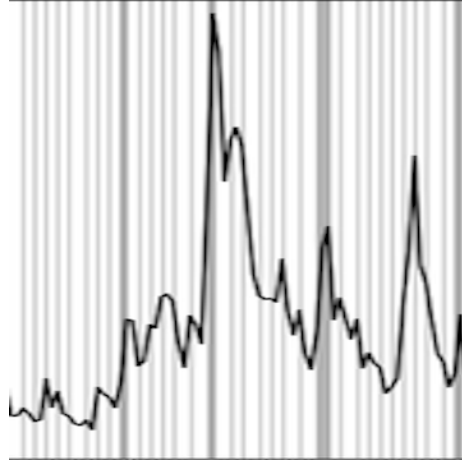


Figure 5: 2007 - 2009 VIX readings

While the model output is a bit difficult to grasp, it does produce accurate readings. This is because even when the value of the VIX drops sharply, it represents a large decrease in implied volatility (IV), which is volatile in nature. The tricky part of interpreting this data is that even though underlying price movements can dip down, depending on the rate and severity, it can still be considered volatile. To assist this understanding, it's important to recognize the factors which are included when calculating implied volatility. IV takes into account many market observations including the underlying equity price, the risk free rate, the strike price of a given option, the time to expiration, and the price of the option itself [2]. With all of this in mind, the reader can start to understand why so many high volatility readings are given from the HMM.

5 Conclusion

This research demonstrates that Hidden Markov Models provide a well-suited framework for regime detection across macroeconomic variables. The inherent structure of HMMs aligns with the nature of economic cycles, where underlying regimes like expansions, contractions, inflationary pressures, and deflationary periods cannot be directly observed but manifest through measurable indicators. By modeling the S&P 500, CPIAUNCS, INDPRO, and VIX, valuable insights were provided by the HMM. The model's ability to probabilistically identify regime transitions proved particularly valuable during historical periods of economic inflection, such as the 2008 financial crisis and the 2020 pandemic shock. This application demonstrates that HMMs allow theoretical constructs to provide practical, actionable insights for investment decision-making, risk management, and economic forecasting by revealing the hidden structure within seemingly noisy macroeconomic data.

At its core HMMs are a robust indicator. However, they do not provide signals for executing individual trades or investments. A further application of this research would be to utilize HMMs as the backbone of a trading strategy, determining which strategies to run based on what regime a given set of variables are currently in.

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